SYSTEMATIC PORTFOLIOS / INDEPENDENT RESEARCH

Does rebalancing travel well?

Entry dates, holding horizons and asset pairs

QuantLab Research

20 October 2026

ASSET PAIRS
11
Ten ETF combinations plus AAPL/TSLA

POLICY-WINDOW RUNS
5,940
Five policies; matched starts and ends

HOLDING HORIZONS
1 / 3 / 5
Calendar years; overlapping windows

The AAPL/TSLA result is not a universal rebalancing law. Weekly rebalancing beats buy and hold in most tested AAPL/TSLA windows, yet several ETF combinations show the opposite result, especially over five years.

Executive findings

A fixed initial 50/50 portfolio is evaluated under daily, weekly, monthly, quarterly and buy-and-hold policies. The analysis retains the original two-stock case and adds every unordered pair from SPY, TLT, GLD, IWM and EFA. No pair is removed after seeing returns, and no frequency or weight is optimized within a window.

Each pair has 60 one-year, 36 three-year and 12 five-year windows. Entry occurs monthly, with the original 6 January 2020 entry retained for the first month. These are heavily overlapping histories, not hundreds of independent trials.

Weekly AAPL/TSLA beats buy and hold on CAGR in 80% of one-year windows, 100% of three-year windows and 91.7% of five-year windows. The three-year result does not imply a 100% future success probability: all 36 observations share much of the same six-year history.

Among the ten ETF pairs, weekly beats buy and hold in all twelve five-year SPY/GLD, GLD/IWM and IWM/EFA windows, but in none of the five-year windows for SPY/TLT, SPY/EFA, TLT/GLD, TLT/IWM or TLT/EFA. Effect sizes range from a few basis points to several annual percentage points. Frequency, horizon and asset pairing all matter.

The defensible conclusion is conditional: rebalancing changes exposure to relative winners and losers, and that exposure change can help or hurt. This study describes the variation; it does not establish an out-of-sample selection rule for choosing favorable pairs.

Research status. Retrospective analysis, not an investment recommendation or a verified trading record. Independent institutional-style presentation; no affiliation with an investment firm. Definitions, execution assumptions and limitations are documented inside.

01 / CONTROLLED DESIGN

1 Fix the rules before comparing windows

The expanded universe is chosen by asset role, not by realized strategy returns.

Symbol	Broad role	Provider reference
SPY	US large-cap equities	State Street SPDR S&P 500 ETF Trust
TLT	Long-duration US Treasury exposure	iShares 20+ Year Treasury Bond ETF
GLD	Gold exposure	SPDR Gold Shares
IWM	US small-cap equities	iShares Russell 2000 ETF
EFA	Developed equities outside US / Canada	iShares MSCI EAFE ETF

The five-ETF universe is documented before downloading its new research results. All ten unordered combinations are retained, including combinations with overlapping equity exposures. AAPL/TSLA is reported separately as the legacy case. This is a retrospective design recorded before these calculations, not a universe chosen in 2019 or an independent historical preregistration.

Matched investment windows

Every comparison uses the same initial USD 100,000, entry date and final valuation date for all five policies. After January 2020, entry is the first exchange session of each month. January 2020 uses 6 January to retain the original convention; 3 January is available as the preceding cash baseline. The window ends at the last session on or before the exact entry-date anniversary after one, three or five calendar years. A window is omitted if that anniversary lies beyond 31 December 2025.

Each window starts fresh. It does not inherit holdings from an earlier entry cohort. No policy is selected using the outcome of that same window, and all recurring policies use a 50/50 strategic target. Calendar resets are known at the preceding close; execution occurs at the next scheduled open under the original integer / cost convention.

Horizon	Entry cohorts per pair	Policies	Total rows across 11 pairs
1 year	60	5	3,300
3 years	36	5	1,980
5 years	12	5	660

No independence shortcut. Monthly cohorts overlap; asset pairs also share instruments and market shocks. The unit counts above describe calculations, not effective statistical sample sizes. Win fractions are not assigned binomial confidence intervals.

02 / FULL-HISTORY CONTEXT

2 Different pairs produce different outcomes

First compare the original six-year holding period, without an entry-date search.

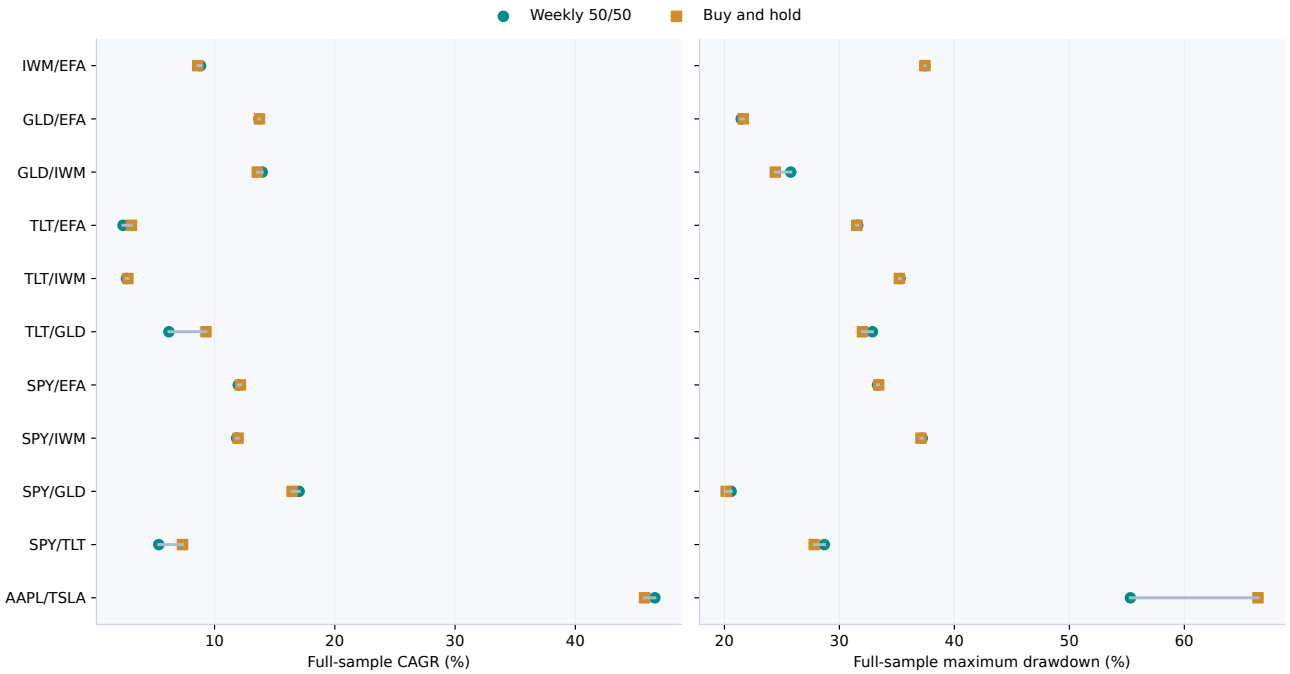


Figure 1: Weekly 50/50 and initial-50/50 buy and hold over 6 January 2020 open–31 December 2025 close. Lower maximum drawdown is preferable. Each line joins the two policies for one fixed pair; axes are not shared across the return and drawdown panels.

The original AAPL/TSLA case has much greater growth and risk than the broad ETF pairs. Its weekly CAGR of 46.61% and drawdown of 55.31% should not be generalized to ordinary multiasset portfolios.

For SPY/TLT, weekly returns 5.36% annually versus 7.34% for buy and hold, while its maximum drawdown is slightly larger. SPY/GLD gives the opposite growth ordering: 17.03% for weekly versus 16.45% for buy and hold, though weekly also has a slightly deeper full-sample drawdown. A growth benefit and a drawdown benefit are separate outcomes.

A lower correlation is not sufficient to guarantee outperformance. Relative growth trends can allow a buy-and-hold portfolio to accumulate a persistent winner, whereas rebalancing keeps restoring exposure to the laggard. This is an interpretation consistent with the observations; a causal decomposition is developed separately in Research Note 04.

03 / HORIZON CHECK

3 A high win fraction can coexist with a small effect

Read the percentage of favorable windows together with the size of the advantage.

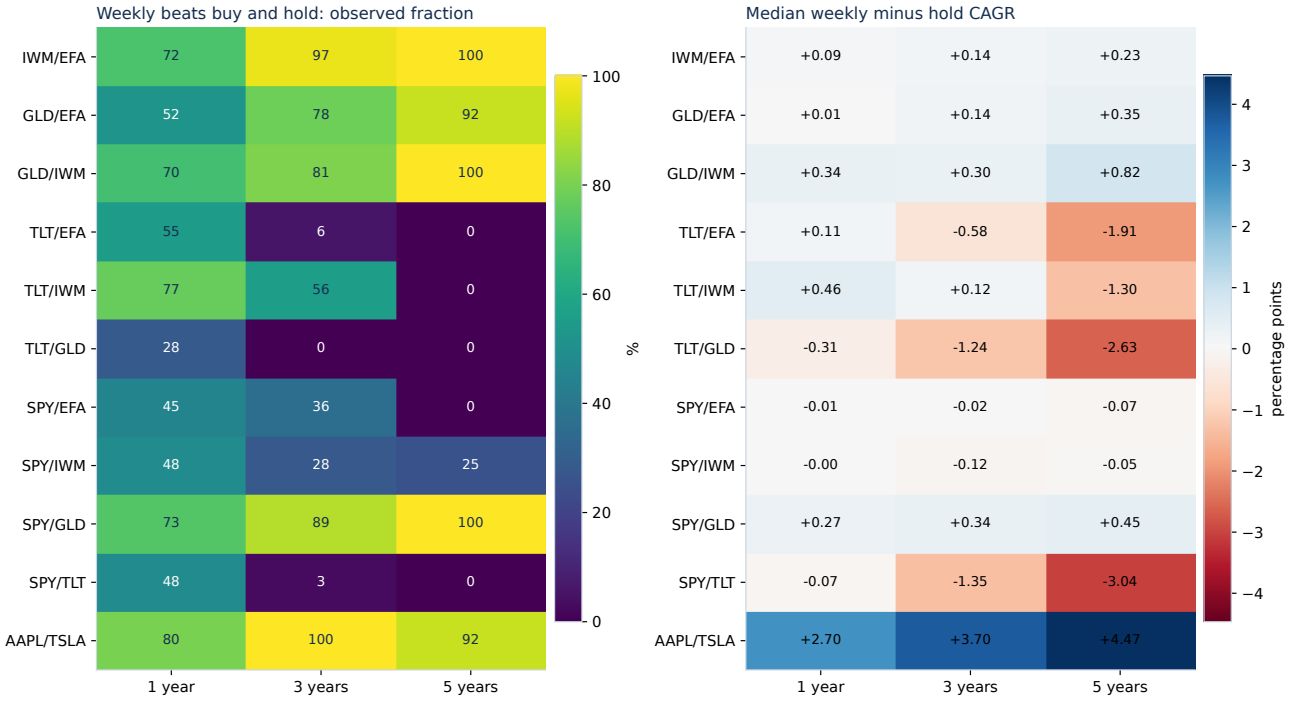


Figure 2: Weekly versus buy-and-hold matched windows. Left: fraction with strictly higher net CAGR. Right: median CAGR difference in percentage points. Counts per cell are 60, 36 and 12 for one-, three- and five-year horizons. Overlap makes these descriptive fractions.

Some comparisons have a very high observed win fraction but only a modest advantage. Weekly IWM/EFA beats buy and hold in all twelve five-year windows, with a median CAGR advantage of about 0.23 percentage points. In contrast, SPY/TLT loses in all twelve five-year windows with a median CAGR disadvantage of about 3.04 points.

AAPL/TSLA's median weekly advantage is 2.70 points for one-year windows, 3.70 for three-year windows and 4.47 for five-year windows. This differs from its smaller full-history CAGR advantage because the cohorts and endpoints differ. It is not an arithmetic inconsistency: a median of rolling-window outcomes is a different statistic from one continuous six-year investment.

The five-year observations all start during 2020 and end during 2025. They cannot establish performance for five-year investments begun in other business cycles. A 100% cell reflects only those closely related paths.

04 / ENTRY-DATE EFFECT

4 The start date changes the story

The research separates rebalance-calendar phase from the date capital is first committed.

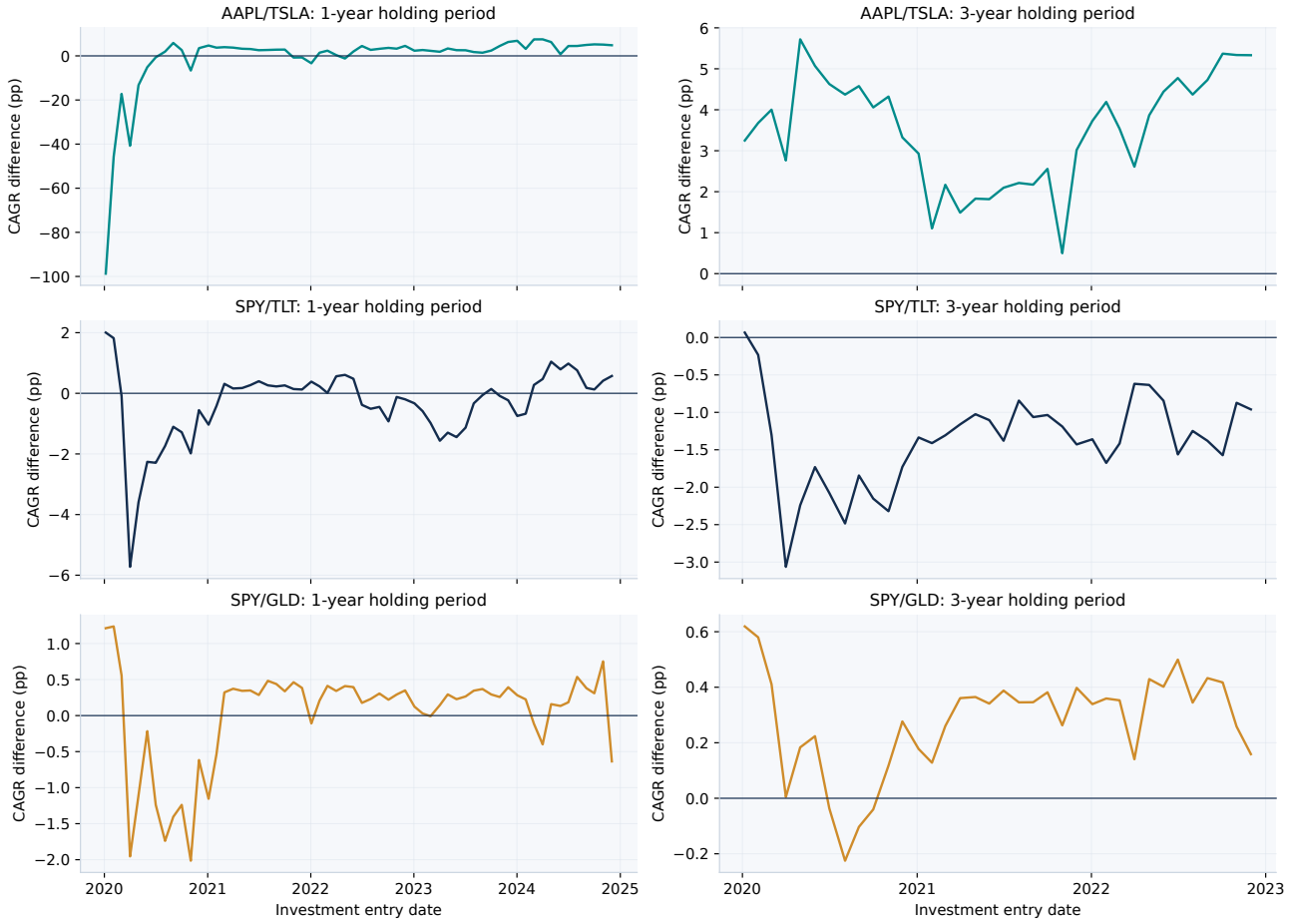


Figure 3: Net CAGR advantage of weekly over buy and hold for three illustrative pairs. One- and three-year horizons use their own matched endpoints. These are entry-cohort comparisons, not portfolio returns earned sequentially through time.

For AAPL/TSLA, weekly's advantage varies substantially with the starting month. Starting near a relative run-up, drawdown or recovery changes how much the buy-and-hold weight drifts and how subsequent reversals affect the two policies.

SPY/TLT illustrates a different history: many three-year entry cohorts favor buy and hold. SPY/GLD more often favors weekly growth, but not in every one-year window. No one of these examples is promoted into a forecasting rule.

The previous frequency-and-weight paper shifted the phase of a rebalance calendar while keeping the investment start fixed. Here, the entry date itself changes, along with the realized investment horizon. These experiments answer different questions and should not be treated as duplicates.

05 / EFFECT DISTRIBUTIONS

5 The typical effect is pair specific

Show the range of cohort outcomes instead of pooling unlike investments.

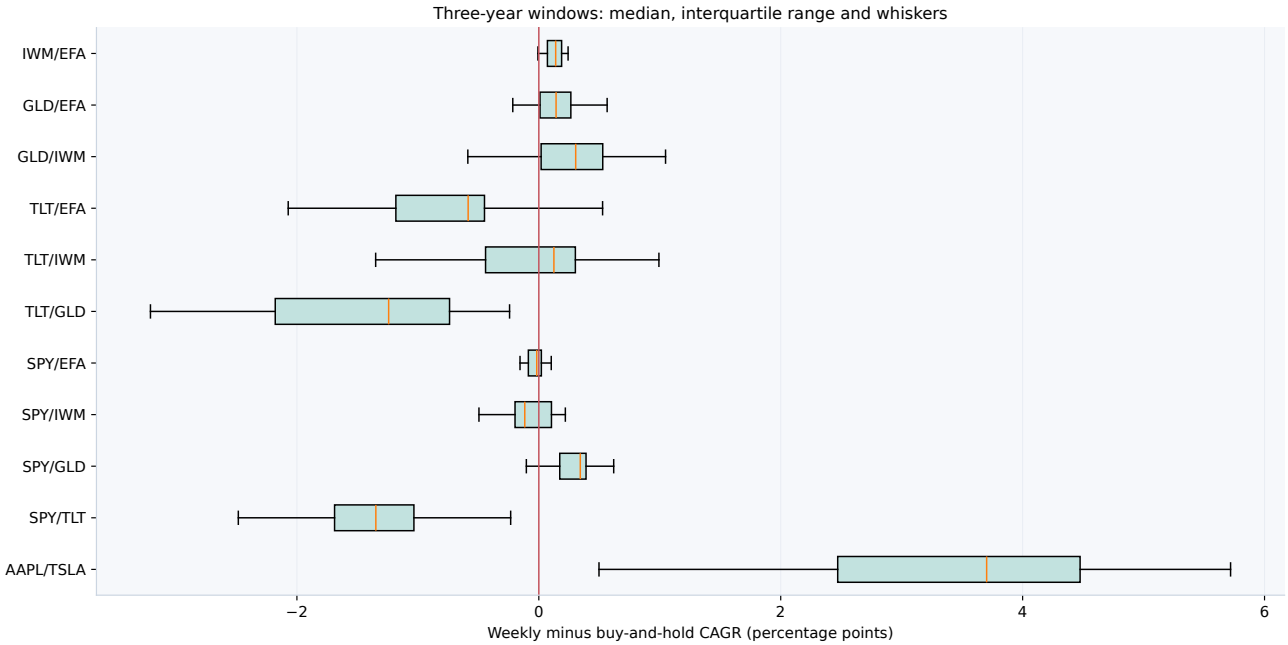


Figure 4: Three-year net CAGR differences across all 36 monthly cohorts per pair. Boxes show medians and interquartile ranges; whiskers use the standard 1.5-IQR rule and outliers are omitted from this view. Full observations remain in the CSV. These are descriptive distributions of overlapping windows.

Three-year pair	Win fraction	Median growth	10th pct.	90th pct.	Median DD
AAPL/TSLA	100.0%	+3.70	+1.82	+5.20	+0.58
SPY/TLT	2.8%	-1.35	-2.20	-0.74	+0.25
SPY/GLD	88.9%	+0.34	-0.02	+0.43	-0.20
SPY/IWM	27.8%	-0.12	-0.37	+0.14	+0.07
SPY/EFA	36.1%	-0.02	-0.11	+0.08	+0.01
TLT/GLD	0.0%	-1.24	-2.61	-0.49	+0.58
TLT/IWM	55.6%	+0.12	-1.23	+0.45	-0.27
TLT/EFA	5.6%	-0.58	-1.55	-0.39	+0.10
GLD/IWM	80.6%	+0.30	-0.12	+0.78	-0.17
GLD/EFA	77.8%	+0.14	-0.14	+0.33	-0.11
IWM/EFA	97.2%	+0.14	+0.03	+0.21	-0.10

All difference columns are percentage points, weekly minus buy and hold; a negative drawdown difference is favorable. The table's 10th and 90th percentiles summarize cohort variability, not confidence limits on expected future performance. Small differences can be sensitive to fees, integer rounding or vendor revisions, and are not assigned statistical significance.

06 / FREQUENCY CROSS-CHECK

6 A weekly label is not a universal solution

Retain all calendar references to see whether a pair result is specific to one schedule.

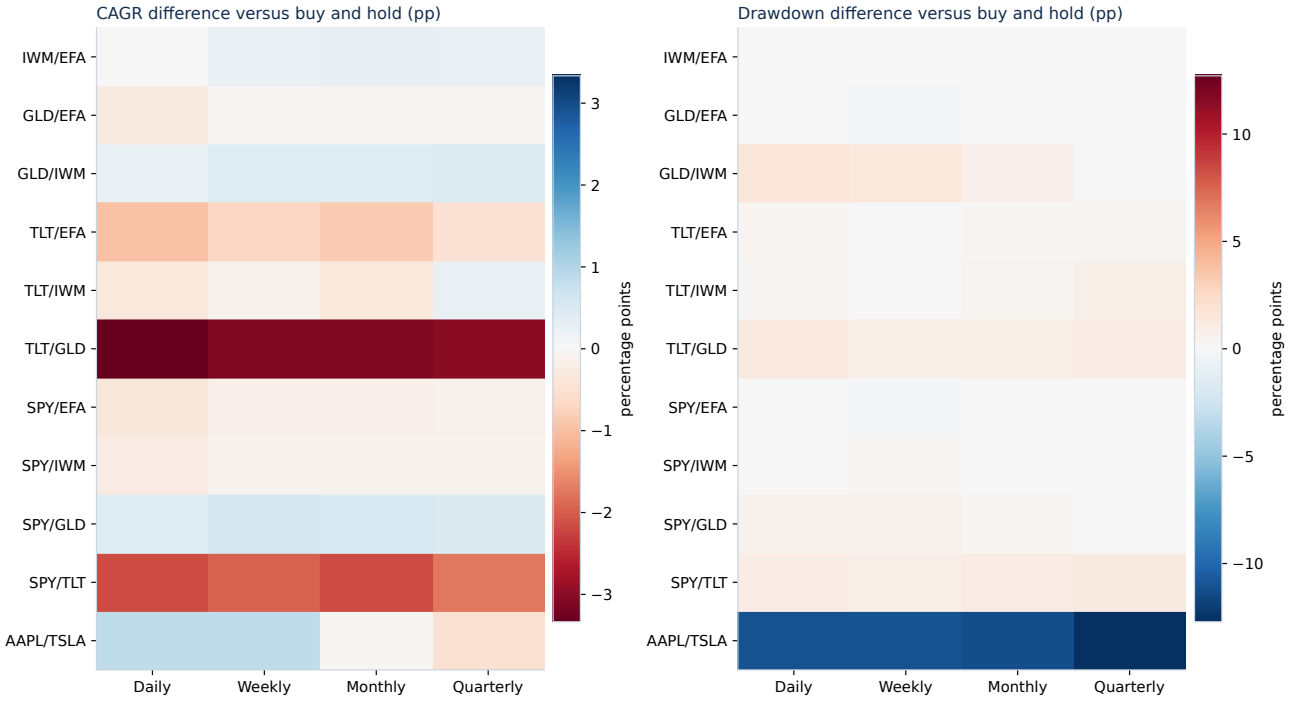


Figure 5: Full-sample differences from buy and hold at four fixed 50/50 rebalance frequencies. Blue is favorable in each panel: higher CAGR on the left and a smaller positive-loss drawdown on the right. Color scales differ by metric and are centered at zero.

The pair-to-pair variation often exceeds the differences among ordinary calendar schedules within a pair. Broadly similar frequency outcomes for one pair do not establish a broad asset-allocation law. Conversely, a frequency change can improve one metric while worsening another.

There is no per-pair optimization in this paper. Choosing the best cell after viewing this matrix would create a new selection problem and would require a new untouched evaluation period. The correct interpretation is conditional robustness of fixed rules, not discovery of a universally superior calendar.

Conclusion. Weekly rebalancing performs well against buy and hold across the tested AAPL/TSLA entry cohorts, but the result does not transfer uniformly to other asset pairs. The evidence supports reporting a distribution across starts and assets rather than presenting one six-year result as representative.

The broader universe also changes the economic question. An equity/bond pair, two regional equity funds and a gold/equity pair need not share the same relative-return dynamics. The five funds were selected as transparent exposure proxies, not as a complete or optimal investment universe.

07 / AUDIT

7 Frozen multiasset evidence

Vendor checks and independent window-boundary checks accompany the result tables.

Data and implementation checks

The legacy dataset is preserved unchanged. Five new ETF histories were downloaded from Yahoo Finance via yfinance with unadjusted raw fields and adjusted close. The 7,535 normalized ETF rows contain 1,507 observations for each fund. All dates match XNYS sessions; checks require unique date/symbol keys, positive OHLCV, no missing required fields and internally consistent OHLC ranges. Raw vendor frames and download metadata are retained.

The new SPY adjusted path, normalized to its first observation, differs from the earlier SPY snapshot by less than 9.05×10^{-7} in relative terms. This is an internal vintage comparison, not independent vendor verification. The original AAPL/TSLA weekly full path still matches the archived equity within 3.50×10^{-10} USD.

Three independently sliced windows reproduce the corresponding batched-window equity paths exactly: weekly SPY/TLT, monthly SPY/GLD and quarterly AAPL/TSLA. All five policies are present for every matched cohort; no result-based filtering is applied.

Accounting and metrics

All historical opening-execution runs use USD 100,000 initial cash, two assets, no leverage or shorting, integer adjusted research units and nonnegative cash. Raw OHLC is scaled by adjusted close / raw close. Dividends and splits are not posted again. Adjusted units preserve the original QuantLab convention but are not historical legal share counts; per-unit commissions inherit this limitation.

Base costs are 5 bps adverse opening slippage and USD max(1, 0.005 times quantity) per executed instrument order. Sales precede buys; symbol order breaks purchase ties. Final holdings are marked without liquidation. Slippage is reflected in fills and is not deducted twice. No taxes, cash interest, capacity constraints or intraday drawdown estimates are included.

CAGR uses elapsed calendar days / 365.25. Annual volatility is the sample daily-return standard deviation times $\sqrt{252}$; Sharpe is the arithmetic mean divided by that standard deviation times $\sqrt{252}$, with zero risk-free rate. MDD is the positive magnitude of the largest decline from a running closing peak. Gross turnover sums opening-reference traded notional / pre-trade opening equity and divides by elapsed years, including entry; it is not divided by two. This turnover convention differs from the original engine's prior-close / fill-notional convention.

Files and reproduction

`data/` freezes market inputs and provenance. `results/` contains complete numerical tables, audit checks and compressed paths where applicable. `code/` contains calculation, plotting and report-generation scripts; `figures/` keeps vector PDFs and PNG previews. A SHA-256 manifest records delivered-file integrity. Run from the project directory in the documented Python environment:

```
python code/research.py
python code/validate.py
python code/plot_figures.py
python code/build_report.py
mkdir -p /tmp/quantlab-paper-build
pdflatex -output-directory=/tmp/quantlab-paper-build report.tex
pdflatex -output-directory=/tmp/quantlab-paper-build report.tex
```

Copy the compiled PDF from the temporary build directory. No network is required to rerun frozen-input calculations. The report builder combines computed tables with fixed English interpretations; changed assumptions require a review of those interpretations.

Primary references and evidence limits

ETF exposure descriptions are drawn from the issuers' fund pages: State Street's SPY and GLD pages, and iShares' TLT, IWM and EFA pages. These descriptions identify the universe; they do not support this paper's strategy-performance claims.

https://www.ssga.com/us/en/individual/etfs/spdr-sp-500-etf-trust-spy

https://www.ssga.com/us/en/individual/etfs/spdr-gold-shares-gld

https://www.ishares.com/us/products/239454/ishares-20-year-treasury-bond-etf

https://www.ishares.com/us/products/239710/ishares-russell-2000-etf

https://www.ishares.com/us/products/239623/ishares-msci-eafe-etf

This universe is retrospective and survivorship conditioned. Only 2020–2025 is observed; longer-history data and post-2025 deployment are not claimed. Overlapping cohorts and shared assets prohibit an independence interpretation of the win fractions. No significance test, future pair-selection rule or universal optimal frequency is asserted.